

TrES-1b

R-1748

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_160816 · created 2026-08-02T19:32:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457616.7722 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.04
Transit depth (Rp/Rs)^2	1.73 +/- 1.06 [%]
Orbital Inclination (inc)	89.81 +/- 1.28
Airmass coefficient 1 (a1)	1.335 +/- 0.046
Airmass coefficient 2 (a2)	-0.2 +/- 0.016
Scatter in the residuals of the lightcurve fit is	3.42 %
Best Comparison Star	None
Optimal Aperture	3.64
Optimal Annulus	9.34
Transit Duration (day)	0.1016 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.131 $\pm$ 0.04 vs 0.1358 (deviation 0.09 σ)
Duration fit vs published	0.1016 d vs 0.1042 d (deviation 0.39 σ)
Mid-time O-C	4.0 min
Depth SNR	2.4
Residual scatter	3.42 %

Light curve

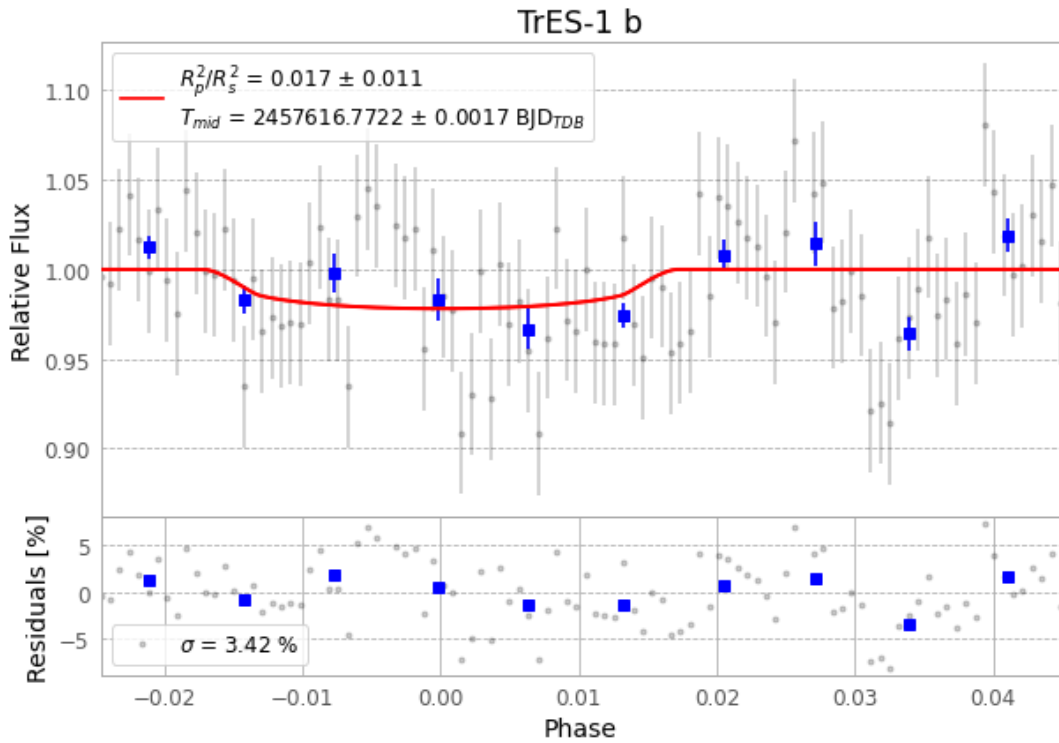


Figure 1 — FinalLightCurve_TrES-1 b_2016-08-15.png (SHA-256 5fe472b22597f18c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 298], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5b818020f8bed959...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

102 FITS frames (67 MB), TRES-1160816043912.FITS → TRES-1160816094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-160816.log (9db7716fc862...), FinalLightCurve_TrES-1 b_2016-08-15.png (5fe472b22597...), FinalLightCurve_TrES-1 b_2016-08-15.csv (f36eb135f29d...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 19:32Z.